

Ballot words and the Morin Thom polynomials

Chern++

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Abstract

The Chern coefficients $C(M)$ of the Morin Thom polynomials Tp_{A_d} are indexed by zero-sum multisets M of d integers. Let $b(M)$ be the number of orderings of M whose proper partial sums are nonnegative. We prove, given two geometric inputs that we state precisely, that $C(M) = b(M)$ whenever $\max M \leq 1$. On that sector $b(M)$ is a Kreweras number. We conjecture that $C(M) \geq b(M)$ for every M , with equality exactly on that sector. This implies Rimányi's positivity conjecture. We prove it, with its equality case, for $d \leq 5$. The case $d = 4$ rests on a new explicit positive product formula for the chamber series together with exact integer-set inclusions. The case $d = 5$ uses a residue-null change of numerator, shown null by a contour-free swap lemma. After it the chamber series is nonnegative, and one coefficient per packet already exceeds the ballot count once the charge is at least 10; the rest is a finite exact computation. The conjecture also holds on every exact coefficient we can compute: all $d \leq 7$ on large boxes, including Tp_{A_7} to $\ell = 5$, Tp_{A_6} to $\ell = 12$ and every packet of charge ≤ 7 at $d = 7$.

1 Setting

By Rimányi's ℓ -free reduction [2] and the Bérczi–Szenes residue formula [1],

$$\text{Tp}_{A_d}^\ell = \sum_M C(M) \prod_{a \in M} c_{\ell+1+a},$$

where M ranges over multisets of d integers summing to 0, $c_0 = 1$, and $c_j = 0$ for $j < 0$. The coefficient $C(M)$ does not depend on ℓ . In chamber coordinates,

$$C(M) = \sum_{\alpha} A_{\beta(\alpha)},$$

where the sum is over the distinct orderings α of M , the prefix sums $\beta_j = \alpha_1 + \cdots + \alpha_j$ ($j < d$) must be nonnegative, and $F_d = \sum_{\beta} A_{\beta} x^{\beta}$ is the Morin chamber series. Rimányi's conjecture is $C(M) \geq 0$.

Definition 1. A *ballot ordering* of M is an ordering whose proper prefix sums are all nonnegative. Let $b(M)$ be the number of distinct ballot orderings. Equivalently, $b(M)$ is $C(M)$ computed with the pure chain series $\prod_{j=1}^{d-1} (1 - x_j)^{-1}$, all of whose coefficients are 1, in place of F_d .

The decreasing ordering is always ballot, so $b(M) \geq 1$. The *charge* of M is $\max M$. The *plane sector* is charge ≤ 1 ; for $\ell = L - 1$ these are the Chern monomials whose largest index is at most $\ell + 2$.

2 The plane sector

Theorem 2. Assume the geometric inputs (I1) and (I3) of §2.1. If $\max M \leq 1$ then $C(M) = b(M)$. Write $M = (1^{d-b}, 1 - s_1, \dots, 1 - s_b)$ with every $s_i \geq 1$, and let m_i be the number of s 's

equal to i . Then this common value is the number of noncrossing partitions of $[d]$ with block sizes $s_1, \dots, s_b$:

$$C(M) = b(M) = \frac{d!}{(d-b+1)! \prod_i m_i!}.$$

Independently of (I1) and (I3), the identity $C(M) = b(M)$ is verified by exact computation from the chamber series on every plane-sector packet with $d \leq 7$.

2.1 The geometric inputs

Fix $n = d + 1$ and $r \geq 2$.

- Let $C_{n,r}$ be the *curvilinear component* of the punctual Hilbert scheme $\text{Hilb}_0^n(\mathbb{C}^r)$. This is the closure of the locus of length- n subschemes supported at the origin and isomorphic to $\text{Spec } \mathbb{C}[t]/(t^n)$. It is irreducible of dimension $(r-1)d$.
- Let V be the tautological rank- n bundle, whose fibre at Z is $\mathcal{O}_Z$.
- Let $E = \ker(V \rightarrow \mathcal{O})$ be its rank- d augmentation part, with Chern roots $\epsilon_1, \dots, \epsilon_d$.

Write the Thom polynomial in the Chern classes $c_i = e_i(x_1, \dots, x_q)$ of formal roots $x_1, \dots, x_q$, where $q = r + \ell$; taking r large enough makes q exceed every index that occurs.

(I1) *Incidence formula.* $\text{Tp}_{A_d}^\ell = \int_{C_{n,r}} \prod_{a=1}^q \prod_{i=1}^d (x_a + \epsilon_i).$

This is the curvilinear form of the Bérczi–Szenes theorem [1]. Reason: a q -tuple of function germs on $\mathbb{C}^r$ defines an A_d singularity exactly when its zero scheme is curvilinear of length n . The incidence variety of pairs (curvilinear Z , q -tuple vanishing on Z) is a vector bundle over $C_{n,r}$, namely the kernel of evaluation onto $E \otimes \mathbb{C}^q$. Its projection to the space of q -tuples is generically one-to-one onto the A_d locus, which has codimension $qd - (r-1)d = d(\ell+1)$. The pushforward is the displayed integral. In exactly this form, with this proof, it is stated in [3].

- (I2) *Extraction.* Let $\lambda = (\lambda_1, \dots, \lambda_s)$ have all parts $\leq q$ and $s \leq d$, and let $\nu = \text{sort}(q^{d-s}, q - \lambda_1, \dots, q - \lambda_s)$. Then the coefficient of the Chern monomial c_λ in (I1) is $\int_{C_{n,r}} m_\nu(\epsilon)$, where m_ν is the monomial symmetric function.

Proof. $\prod_i \prod_a (x_a + \epsilon_i) = \prod_i \sum_{j=0}^q e_j(x) \epsilon_i^{q-j}$. The coefficient of $e_{\lambda_1}(x) \cdots e_{\lambda_s}(x) e_0(x)^{d-s}$ is the sum over distinct assignments of the indices $\lambda_1, \dots, \lambda_s, 0^{d-s}$ to the d roots, that is, $m_\nu(\epsilon)$. $\square$

- (I3) *Hyperplane lemma.* $c_d(E) \cap [C_{n,r}] = [C_{n,r-1}]$, where $C_{n,r-1}$ is embedded as the subschemes lying in a coordinate hyperplane.

Reason: the last coordinate u_r gives a section of E . It vanishes exactly on the subschemes contained in $\{u_r = 0\}$. On the standard curvilinear chart $I = (u_2 - p_2(t), \dots, u_r - p_r(t), t^n)$ it is the regular d -tuple of coefficients of p_r , so the zero locus is reduced of the expected codimension d . This is stated and proved in this form in [3].

- (I4) *The plane (classical).* For $r = 2$, $C_{n,2} = \text{Hilb}_0^n(\mathbb{C}^2)$ by Briançon's theorem [4]. Its cohomology is identified by Lehn–Sorger [5] with the associated graded of the class algebra of S_n (filtered by n minus the number of cycles). Under this identification the elementary symmetric functions of the ϵ_i go to those of the Jucys–Murphy elements $J_k = \sum_{a < k} (a k)$. The top degree is spanned by the class of the n -cycles, normalised by $\int c_d(E) = 1$. So for $|\nu| = d$, $\int_{C_{n,2}} m_\nu(\epsilon) = [\sigma] m_\nu(J_1, \dots, J_n)$ for any n -cycle σ .

Reduction to the plane. Take $\ell = L - 1$ and a plane-sector M , so the Chern monomial is c_λ with $\lambda_i = L + \alpha_i \leq L + 1$ for the entries α_i of M with $L + \alpha_i > 0$. With $q = r + L - 1$, every entry of ν is at least $q - L - 1 = r - 2$. While $r > 2$ all entries are positive, so $m_\nu(\epsilon) = (\epsilon_1 \cdots \epsilon_d) m_{\nu-1}(\epsilon) = c_d(E) m_{\nu-1}(\epsilon)$. By (I3) this lowers r by one and every entry of ν by one. At $r = 2$, $q = L + 1$ and the entries of ν are $L + 1 - \lambda_i = 1 - \alpha_i$, together with $q = 1 - (-L)$ for the entries $\alpha = -L$ that give c_0 . So ν is the multiset $1 - M$, and by (I4)

$$C(M) = [\sigma] m_{1-M}(J_1, \dots, J_n).$$

Everything below is new.

Lemma 3. *Let $\sigma(x) = x + 1 \pmod{n}$. A minimal monotone factorisation of σ is an expression $\sigma = (a_1 b_1)(a_2 b_2) \cdots (a_{n-1} b_{n-1})$ with $a_i < b_i$ and $b_1 \leq \cdots \leq b_{n-1}$. Its exponent vector is $e_k = \#\{i : b_i = k\}$, with $e_1 = 0$. Then for every $e \in \mathbb{Z}_{\geq 0}^n$ with $\sum e_k = n - 1$:*

$$\#\{\text{minimal monotone factorisations of } \sigma \text{ with exponent vector } e\} = \begin{cases} 1 & \text{if } D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0 \text{ for all } j \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Necessity. The $n - 1$ transpositions of a minimal factorisation of an n -cycle are the edges of a tree on $[n]$. Those with $b_i \leq j$ are edges of a forest on $[j]$, so there are at most $j - 1$ of them.

Existence and uniqueness, by induction on n . The case $n = 1$ is trivial. Let $m = e_n$ and let $(a_1 n) \cdots (a_m n)$ be the final m factors, with product C . The remaining factors form a minimal monotone factorisation of $F = \sigma C^{-1}$, a permutation of $[n - 1]$ with exactly m cycles. Since $C^{-1} = (n a_1 a_2 \cdots a_m)$ we have $a_1 = \sigma^{-1}(n) = n - 1$ and

$$F(x) = x + 1 \ (x \notin \{a_i\}), \quad F(a_i) = a_{i+1} + 1 \ (i < m), \quad F(a_m) = 1.$$

Because $F \cdot C = \sigma$ is one cycle and C is an $(m + 1)$ -cycle, each cycle of F contains exactly one a_i . Starting at $a_{i+1} + 1$ and stepping by $+1$, the cycle through a_i must meet a_i before any other a_j . So $a_{i+1} < a_i$ with no a_j in between, and a_m is the least. Hence $a_1 > a_2 > \cdots > a_m$, and F is the product of the increasing cycles on the consecutive intervals

$$I_m = [1, a_m], \quad I_{m-1} = [a_m + 1, a_{m-1}], \quad \dots, \quad I_1 = [a_2 + 1, n - 1].$$

Transpositions on disjoint intervals commute, so monotone factorisations of F are exactly shuffles, merged by b , of monotone factorisations of the interval cycles. Minimality on each interval says the exponents inside I_{m-i} sum to $|I_{m-i}| - 1$. Since D changes by $1 - e_k \leq 1$ per step, this means $D(a_{m-i}) = i$, and the ballot condition inside the interval means $D > i$ strictly after a_{m-i} , up to the next cut. So a_{m-i} is the last j with $D(j) = i$. When e is ballot these last visits exist, are increasing in i , and end at $D(n - 1) = m - 1$. So the cut points exist and are forced, and the induction applies to each interval (each is again an increasing cycle, of smaller length). $\square$

Proof of Theorem 2. Expanding $m_\nu(J_1, \dots, J_n) = \sum_e \prod_k J_k^{e_k}$ over distinct rearrangements e of ν (padded by zeros, $e_1 = 0$ since $J_1 = 0$), the coefficient of σ counts pairs $(e, \text{minimal monotone factorisation with exponent vector } e)$. By Lemma 3 this is the number of ballot vectors e . Setting $\alpha_k = 1 - e_{k+1}$ ($k = 1, \dots, d$) turns the condition $D \geq 0$ into $\alpha_1 + \cdots + \alpha_j \geq 0$, and e into an ordering of M . So $C(M) = b(M)$. For the product formula, apply the cycle lemma to $-M$ with a final -1 appended: n integers ≥ -1 summing to -1 , exactly one cyclic rotation of which has all proper prefix sums nonnegative. Reversal identifies these rotations with ballot orderings of M , so $b(M) = \frac{1}{n} \cdot \frac{n!}{(d-b+1)! \prod_i m_i!}$. This is Kreweras' count [6]. $\square$

Lemma 3 is checked exhaustively for $n \leq 7$; the totals are the Catalan numbers 1, 2, 5, 14, 42, 132. The identity $C(M) = b(M)$ is checked on every plane-sector packet for $d \leq 7$ directly from the chamber series. Both checks are in tier 12 of the test suite.

Remark 4. On the plane sector the Morin chamber series is therefore indistinguishable, packet by packet, from the pure chain series $\prod_j (1 - x_j)^{-1}$. For example, at $d = 7$ the packet $M = (1, 1, 1, 0, -1, -1, -1)$ has 35 ballot orderings. Eight of them carry nonzero A_β , namely 1, 4, -1, -4, 5, 6, 6, 18, and these sum to 35.

3 The ballot conjecture

Conjecture 5 (ballot). *For every zero-sum multiset M of d integers, $C(M) \geq b(M)$, with equality if and only if $\max M \leq 1$. Equivalently: the series $F_d - \prod_j (1 - x_j)^{-1}$ has nonnegative packet sums, and they vanish exactly on the plane sector.*

Corollary 6. *Conjecture 5 implies Rimányi’s conjecture, and more: every Chern coefficient of Tp_{A_d} is at least 1 whenever the monomial occurs.*

This explains a feature of the data that had no explanation: across $d = 4, \dots, 7$ the least Chern coefficient is 1, never 0. It is attained exactly at $M = 0$ and $M = (1^{d-1}, 1 - d)$, the two multisets with $b(M) = 1$ on the plane sector.

Table 1: The ballot conjecture on every exact table. A violation is $C(M) < b(M)$; the equality columns test that $C = b$ holds exactly on the plane sector.

d	box	packets	violations	plane $C \neq b$	off-plane $C = b$	$\min_{\max M \geq 2} C/b$
2	level box L=14	15	0	0	0	2
3	level box L=14	169	0	0	0	2
4	level box L=12	1033	0	0	0	9/4
5	level box L=12	7166	0	0	0	12/5
6	level box L=11	29941	0	0	0	79/30
7	level box L=6	8033	0	0	0	58/21
7	charge box p=6	8033	0	0	0	58/21
6	level L=13	62239	0	0	0	79/30
7	charge p=7	16475	0	0	0	58/21

Evidence. Table 1 tests the conjecture on every exact table available. The data come from the exact sweep of `chernpp.bboxes`, reconstructed by CRT and verified against a held-back prime. The level box $\min M \geq -L$ contains every packet of Tp^ℓ with $\ell < L$; the charge box contains every packet with $\max M \leq p$. There is no violation, no plane-sector packet with $C \neq b$, and no off-plane packet with $C = b$. Off the plane sector $C(M)/b(M) \geq 2$ throughout. The minimum is 2 at $(2, -1, -1)$ and grows slowly with d (9/4, 12/5, 79/30, 58/21), attained on the family $(2, 1^{d-4}, -1^{d-2})$ and its neighbours.

What does not work.

- *Factorial weights.* $C(p, -1^p) = p!$, which suggests weighting each ballot word by $\prod_{\alpha_i > 0} \alpha_i!$. That bound fails already at $d = 3$.
- *Fusion.* Replacing a 2 by two 1s and counting plane-sector ballot words of the longer multiset gives some families exactly. For example, $C(2, 0^{d-2}, -2) = \binom{d+1}{3} + \binom{d}{2}$, where both terms are plane-sector counts. But the residual is negative elsewhere.

- *Local step weights.* Keep weight 1 on steps ≤ 1 and give each step of size 2 a weight depending on local data: its height, together with the neighbouring steps, the number of remaining steps, or the next return. Fitted exactly on the packets with one 2, every such model that is consistent within a single d is inconsistent across d .
- *Formal Jucys–Murphy extension.* Theorem 2 reads $C(M) = \sum_{\alpha} [\sigma] \prod_k J_{k+1}^{1-\alpha_k}$ over ballot α . At charge 2 this needs J_k^{-1} , which exists for $k = 2, 3$ (their eigenvalues, the contents of boxes 2 and 3, are nonzero). Where it can be evaluated, it returns the wrong values, some of them fractional ($C(2, 0, -2) = 7$ against $23/2$). At $d = 4$ the needed inverses do not exist.
- *Individual coefficients.* No bound on single A_{β} can work: they are unbounded below (-3622 at $d = 7$), and the smallest symmetrisation that clears all negatives grows with d [7].

4 The conjecture for $d \leq 4$

At fixed d the plane sector is finite: its entries lie in $[1-d, 1]$. So the equality part of Conjecture 5 at fixed d is a finite check. It needs neither Theorem 2 nor its geometric inputs. The content is the inequality over infinitely many M , together with strictness off the plane sector. We prove both for $d \leq 3$, directly from the chamber series.

In chamber coordinates write $x = x_1$, $y = x_2$. The chamber series is a power series in these variables, and its exponents $\beta \geq 0$ are exactly the prefix-sum vectors of ballot orderings. For $d \leq 3$ the multidegree numerator is trivial, and the Bérczi–Szenes formula [1] gives

$$F_2 = \frac{1-x}{1-2x}, \quad F_3 = \frac{(1-x)(1-y)(1-xy)}{(1-2x)(1-2xy)(1-y-xy)}.$$

These are the stored chamber algebras of this project, checked in tier 15. The factorisation of the numerator of F_3 is a direct expansion.

Theorem 7. *For $d \leq 3$ and every zero-sum M : $C(M) \geq b(M)$, with equality if and only if $\max M \leq 1$. Moreover $C(M) \geq b(M) + 1$ whenever $\max M \geq 2$.*

Proof. $d = 2$. $F_2 = 1 + x/(1-2x)$, so $C(k, -k) = 2^{k-1}$ for $k \geq 1$ and $C(0, 0) = 1$, while $b = 1$ throughout.

$d = 3$. Each factor of F_3 has the form $\frac{1-u}{1-u-v} = 1 + \frac{v}{1-u-v}$ with u, v monomials with nonnegative coefficients:

$$F_3 = (1+U)(1+V)(1+W), \quad U = \frac{x}{1-2x}, \quad V = \frac{xy}{1-2xy}, \quad W = \frac{xy}{1-y-xy}.$$

All of U, V, W have nonnegative coefficients, so $F_3 \geq 1 + U + V + W + UV + UW$ coefficientwise. Compare with

$$Z_3 = 1 + \frac{x}{(1-x)(1-y)} + \frac{xy}{1-xy},$$

whose coefficient at $x^i y^j$ is 1 at $(0, 0)$, 0 for $i = 0 < j$, and $1 + [i = j]$ for $i \geq 1$.

(i) $V \geq xy/(1-xy)$, since $2^{k-1} \geq 1$.

(ii) $U + W + UW \geq x/((1-x)(1-y))$. Write $[x^i y^j]$ for a coefficient. Then $[x^i]U = 2^{i-1}$ covers $j = 0$. $W = \sum_{m \geq 0} xy(y+xy)^m$ has $[x^{1+k}y^{1+m}]W = \binom{m}{k}$, which covers $1 \leq i \leq j$. And $[x^i y^j]UW \geq [x^{i-1}]U \cdot [xy^j]W = 2^{i-2}$ covers $1 \leq j < i$.

So $F_3 \geq Z_3$ coefficientwise. Next, the packet sums of Z_3 are $b(M)$. A ballot ordering α of $M = (a, b, c)$ has $\beta = (\alpha_1, \alpha_1 + \alpha_2)$. - It lands on the zero set $\{i = 0 < j\}$ exactly when $\alpha = (0, k, -k)$, which requires $M = (k, 0, -k)$. - It lands on the diagonal $i = j \geq 1$ exactly when $\alpha = (k, 0, -k)$, with the same M .

Both occur exactly for $M = (k, 0, -k)$, $k \geq 1$, and then they cancel. So the Z_3 -packet sum is the number of ballot orderings, and $C(M) \geq b(M)$.

For strictness, every cell with $i \geq 2$ receives at least one unit beyond what was used to dominate Z_3 .

- $j = 0$: $[x^i]U = 2^{i-1} \geq 2$.
- $1 \leq j < i$: the unused product UV gives $[x^{i-j}]U \cdot [(xy)^j]V \geq 1$.
- $j \geq i$: (ii) used only W there. The term of UW with x taken from U , namely $[x]U \cdot [x^{i-1}y^j]W = \binom{j-1}{i-2} \geq 1$, is unused.

If $\max M \geq 2$, the decreasing ordering of M is ballot with $\beta_1 = \max M \geq 2$. So $C(M) \geq b(M) + 1$. Finally, the plane sector at $d = 3$ is $\{(0, 0, 0), (1, 0, -1), (1, 1, -2)\}$, where $C = 1, 3, 1 = b$. $\square$

4.1 An explicit positive formula for F_4

The same shape of argument needs a manifestly positive expression for the chamber series. At $d = 4$ one exists. In chamber variables x_1, x_2, x_3 put $s = x_2x_3$ and $t = x_1x_2x_3$. The Morin chamber series is

$$F_4 = \frac{N_4}{(1-2x_1)(1-2x_1x_2)(1-x_2-x_1x_2)(1-2t)(1-s-t)(1-x_3-t)(1-2s)},$$

and its numerator factors completely:

$$N_4 = (1-x_1)(1-x_2)(1-x_1x_2)(1-x_3)(1-s)(1-t)(1-s-2t).$$

Lemma 8.

$$F_4 = \left(\frac{1}{2} \frac{1}{1-2x_1} + \frac{1}{2} \frac{1}{1-2s} + \frac{x_1^2 s}{(1-2x_1)(1-s-t)} \right) \cdot \frac{1-x_1x_2}{1-2x_1x_2} \cdot \frac{1-x_2}{1-x_2-x_1x_2} \cdot \frac{1-t}{1-2t} \cdot \frac{1-x_3}{1-x_3-t}.$$

Each of the last four factors is $1 + v/(1-u-v)$ with u, v nonnegative, and the first factor is a sum of nonnegative series. In particular F_4 has nonnegative coefficients (strong positivity at $d = 4$).

Proof. Group the numerator factors $(1-x_1)(1-s)(1-s-2t)$ with the denominator factors $(1-2x_1)(1-s-t)(1-2s)$, and write $a = x_1$, so that $t = as$. Then

$$(1-a)(1-s)(1-s-2as) = a^2s(1-2s) + \frac{1}{2}(1-2a)(1-s-as) + \frac{1}{2}(1-s-as)(1-2s),$$

which is checked by expanding both sides. Dividing by $(1-2a)(1-s-as)(1-2s)$ gives the first factor, and the remaining four numerator factors pair with the remaining four denominator factors as displayed. $\square$

This expression was found by a small exact certificate search, and both identities are pinned in tier 15.

Theorem 9. *The conclusion of Theorem 7 holds for $d = 4$: $C(M) \geq b(M)$ for every zero-sum M , with equality if and only if $\max M \leq 1$.*

Proof. Step 1: a lower bound for F_4 by supports. Since $\frac{1}{2} \frac{1}{1-2y} = \frac{1}{2} + \frac{y}{1-2y}$, Lemma 8 reads $F_4 = (1 + X + S + Y) \prod_{i=1}^4 (1 + V_i)$ with

$$X = \frac{x_1}{1-2x_1}, S = \frac{s}{1-2s}, Y = \frac{x_1^2 s}{(1-2x_1)(1-s-t)}, V_1 = \frac{x_1x_2}{1-2x_1x_2}, V_2 = \frac{x_1x_2}{1-x_2-x_1x_2}, V_3 = \frac{t}{1-2t}, V_4 = \frac{t}{1-x_3-t}.$$

Each factor has nonnegative integer coefficients, positive exactly on the following supports. Write (a, b, c) for an exponent vector. - X : $b = c = 0 < a$. - S : $a = 0 < b = c$. - Y : $a \geq 2$,

$b = c \geq 1$. - V_1 : $a = b \geq 1, c = 0$. - V_2 : $1 \leq a \leq b, c = 0$. - V_3 : $a = b = c \geq 1$. - V_4 : $1 \leq a = b \leq c$.

The coefficients are powers of 2 or binomial coefficients $\binom{m}{j}$ with $0 \leq j \leq m$. For example, $V_2 = \sum_m x_1 x_2 (x_2 + x_1 x_2)^m$ has coefficient $\binom{m}{j}$ at $(1 + j, 1 + m, 0)$. Expanding the product, F_4 is a sum of 64 series with nonnegative integer coefficients, each positive exactly on a Minkowski sum of these supports. Hence $F_4(\beta) \geq N(\beta)$, the number of those 64 supports that contain β .

Step 2: the comparison series. The zero set of F_4 is $\{a = 0, b \neq c\} \cup \{b = 0 < a, c\}$. Let

$$T = \{b = c \geq 1 \leq a\} \cup \{a = b \geq 1, c = 0\} \cup \{b = a + c, a, c \geq 1\}.$$

The following map ϕ rearranges the word $\alpha = (a, b - a, c - b, -c)$ of a cell of the zero set. It is defined in four pieces:

cells of the zero set	word $\mapsto$ rearranged word	image
$b = 0, a, c \geq 1$	$(a, -a, c, -c) \mapsto (a, c, -a, -c)$	$\{b = a + c\}$
$a = 0, b, c \geq 1, b \neq c$	$(0, u, v, w) \mapsto (u, v, 0, w)$	$\{b = c\}, a \neq b$
$a = b = 0, c \geq 1$	$(0, 0, c, -c) \mapsto (c, 0, 0, -c)$	$\{a = b = c\}$
$a = c = 0, b \geq 1$	$(0, b, -b, 0) \mapsto (b, 0, -b, 0)$	$\{a = b, c = 0\}$

Every piece permutes the entries of α , so ϕ preserves the packet, and its images are ballot. The pieces partition the zero set, each piece is injective, and their images partition T . All of this is decided exactly by `isl (chernpp.ballot.verify_d4_bijection)`. So $Z_4 = P_4 - 1_{\text{zero set}} + 1_T$, where $P_4 = \prod_j (1 - x_j)^{-1}$, has packet sums exactly $b(M)$. Its coefficient is $w = 1_{W_1} + 1_T$, where W_1 is the complement of the zero set.

Step 3: exact inclusions. Let D be the set of cells whose word is weakly decreasing with $\alpha_1 \geq 2$. Every packet with $\max M \geq 2$ has its decreasing ordering in D . The functions N and $w + 1_D \leq 3$ are defined by Presburger formulas. The inequality $N \geq w + 1_D$ is equivalent to three inclusions: for $k = 1, 2, 3$, $\{w + 1_D \geq k\}$ lies inside the set of cells covered by at least k of the 64 supports. These inclusions, and $T, D \subseteq W_1$, are decided exactly by the integer-set library `isl (chernpp.ballot.verify_d4, one second)`, with no truncation.

Summing over a packet: $C(M) \geq \sum N \geq \sum (w + 1_D) = b(M) + [\max M \geq 2]$. The plane sector at $d = 4$ consists of five packets, where $C = b$ is checked directly. $\square$

At $d = 5$ the zero set of a nonnegative representative is far more intricate (for the representative below, 2305 of the 31104 interior cells of one box), so we use a different endgame: a single coefficient already beats $b(M)$ once the charge is large.

5 The conjecture for $d = 5$

The canonical F_5 has negative coefficients, so no argument of the kind above can start from it. We first change the numerator of the residue formula by a kernel that does not change any packet sum, then show that the new series is nonnegative, and finally let its dominant coefficient do the work.

5.1 Chamber expansions and a swap lemma

In the coordinates $z_1, \dots, z_d$ of the residue formula (so that $x_j = z_j/z_{j+1}$), write D_d for the product of the linear forms of the denominator and $V = \prod_{i < j} (z_j - z_i)$. Each factor of D_d has the shape $L = cz_k + \lambda(z_1, \dots, z_{k-1})$ with $c \neq 0$; call z_k its *top variable*. Its inverse has the *chamber expansion*

$$\text{Exp}(1/L) = (cz_k)^{-1} \sum_{n \geq 0} (-\lambda/(cz_k))^n,$$

a series in the monomials z_a/z_b with $a < b$. These monomials generate a pointed monoid Γ , so series supported on translates of Γ form a ring, and Exp extends to a ring homomorphism from polynomials with such factors inverted. For a numerator P , the chamber series $F(P)$ of the residue formula is, up to one sign that depends only on d , $\text{Exp}(PV/D_d)$, written in the variables x_j . There is no monomial factor, since PV/D_d has degree 0. At $d = 5$ this is checked as a polynomial identity for the stored $\mathcal{Q}_5$ (`chernpp.ballot.d5_normalisation`). The packet sum of $F(P)$ at M is $\pm \sum_{\alpha \in S_d M} [z^\alpha] \text{Exp}(PV/D_d)$. Non-ballot orderings contribute 0, since $F(P)$ is a power series in the x_j .

Lemma 10 (swap). *Let $s = (i \ i+1)$, and let $R = N / \prod_r L_r$ with N a polynomial and each L_r a linear form as above. Suppose that $sR = R$ as rational functions, and that R is s -safe: no L_r with top variable z_{i+1} involves z_i . Then $\sum_{\alpha \in O} [z^\alpha] \text{Exp}(VR) = 0$ for every S_d -orbit O of exponents.*

Proof. Let $\Gamma_s \subset \Gamma$ be the monoid generated by the z_a/z_b with $a < b$ and $(a, b) \neq (i, i+1)$. The transposition s permutes these generators, so it acts by ring automorphisms on series supported on translates of Γ_s . If L is safe, then $\text{Exp}(1/L)$ is such a series, and $s \text{Exp}(1/L) = \text{Exp}(1/sL)$.

To see this, apply s termwise and consider the top variable z_k of L .

- $k > i+1$: sL has the same top variable and the lower part $s\lambda$.
- $k = i+1$: safety says $L = cz_{i+1} + \lambda(z_1, \dots, z_{i-1})$, and $sL = cz_i + \lambda$ has top variable z_i .
- $k = i$: $sL = cz_{i+1} + \lambda$ has top variable z_{i+1} .
- $k < i$: $sL = L$.

In every case sL is again safe. Because Exp is multiplicative, $s \text{Exp}(R) = \text{Exp}(sR) = \text{Exp}(R)$. Since $sV = -V$, the coefficients $c(\alpha) = [z^\alpha] \text{Exp}(VR)$ satisfy $c(s\alpha) = -c(\alpha)$. An orbit O is s -stable, so $\sum_O c = \sum_O c \circ s = -\sum_O c$, and the sum vanishes. $\square$

The lemma is the formal counterpart of exchanging two adjacent contours of the iterated residue [3]. Safety is the condition that no pole lies between them. In this form the lemma needs no analysis, and its hypotheses are finite checks.

5.2 A residue-null kernel

Put, as in [3, §4.3],

$$\begin{aligned} B &= 2z_1 - z_2, & F &= 2z_1 - z_3, & E &= z_1 + z_2 - z_3, & D &= 2z_2 - z_4, \\ M &= z_2 + z_3 - z_5, & C &= z_1 + z_4 - z_5, & G &= 2z_1 + z_2 - z_5, & J &= 2z_1 + z_3 - z_5, \end{aligned}$$

and let $P(t) = \prod_{a \in S} (a - t)$ with $S = \{2z_1, z_1 + z_2, 2z_2, z_1 + z_3\}$.

Proposition 11. $\mathcal{Q}_5 = GDJ + GBC$, and the kernel GBC is residue-null: $F(GBC)$ has every packet sum equal to 0. Hence $C(M)$ is the packet sum of the chamber series $F_5^{GDJ} := F(GDJ)$.

Proof. Three identities are checked exactly (`chernpp.ballot.d5_nullity_checks`): the factorisation $\mathcal{Q}_5 = G(DJ + BC)$, the denominator $D_5 = \pm BFE P(z_4)P(z_5)MC$, and $G = F + M$. Hence

$$\frac{GBC}{D_5} = \frac{BMC}{D_5} + \frac{FBC}{D_5} = \pm \frac{1}{FEP(z_4)P(z_5)} \pm H, \quad H = \frac{1}{EP(z_4)P(z_5)M(z_5)},$$

where $M(t) = z_2 + z_3 - t$, and we apply Lemma 10 three times.

- $1/(FEP(z_4)P(z_5))$ is invariant under $s_4 = (45)$. Its only factors with top variable z_5 are $a - z_5$ with $a \in S$, which do not involve z_4 .
- $H + s_4 H = (M(z_4) + M(z_5))/(EP(z_4)P(z_5)M(z_4)M(z_5))$ is s_4 -invariant and s_4 -safe, because $M(z_5)$ does not involve z_4 .

- $H - s_4 H = (M(z_4) - M(z_5))/(\cdots) = (z_5 - z_4)/(ET(z_4)T(z_5))$, where $T(t) = P(t)M(t) = \prod_{a \in S'}(a - t)$ and $S' = S \cup \{z_2 + z_3\}$. The set S' and the form E are invariant under $s_1 = (12)$, so this term is s_1 -invariant. No factor has top variable z_2 , so it is s_1 -safe.

The invariances and safety conditions are also checked mechanically. As negative controls, D_5 itself is not s_4 -safe and FBC/D_5 is not s_4 -invariant. $\square$

The factorisation of $\mathcal{Q}_5$ and the two-swap strategy are due to [3, §§4.3–4.4], which argues with nested contours. The contour-free Lemma 10 is ours.

5.3 The gauged series is nonnegative

In chamber variables put $p = x_1 x_2 x_3 x_4$, $q = x_2 x_3 x_4$, $r = x_3 x_4$ and

$$A = 1 - 2x_1, \quad S = 1 - 2x_1 x_2, \quad P = 1 - 2p, \quad Q = 1 - 2q, \quad U = 1 - r - p, \quad V' = 1 - r - q.$$

Lemma 12. $F_5^{GDJ} = \mathcal{C} \cdot \prod_{i=1}^6 (1 + V_i)$, where

$$\begin{aligned} 1 + V_1 &= \frac{1 - x_2}{1 - x_2 - x_1 x_2}, & 1 + V_2 &= \frac{1 - x_2 x_3}{1 - x_2 x_3 - x_1 x_2 x_3}, & 1 + V_3 &= \frac{1 - x_3}{1 - x_3 - x_1 x_2 x_3}, \\ 1 + V_4 &= \frac{1 - x_4}{1 - x_4 - p}, & 1 + V_5 &= \frac{1 - p}{1 - p - q}, & 1 + V_6 &= \frac{1 - x_1 x_2 x_3}{1 - 2x_1 x_2 x_3}, \end{aligned}$$

and

$$\begin{aligned} \mathcal{C} &= \frac{(1 - x_1)(1 - x_1 x_2)(1 - r)(1 - q)(1 - r - 2p)(1 - q - 2p)}{A S P Q U V'} \\ &= 1 + \frac{x_1}{A} + \frac{x_1 x_2}{S} + \frac{x_1^2 x_2}{A S} + \frac{x_2 r + x_2 r^2}{Q} + \frac{x_1 x_2^2 r}{S Q} + \frac{x_1^2 x_2 r}{A U} + \frac{x_1^3 x_2^2 r}{A S U} + \frac{x_1^2 x_2^3 r^2}{S Q U} + \frac{x_1 x_2^2 r^3}{P V'} + \frac{x_2 r^3}{Q V'}. \end{aligned}$$

In particular F_5^{GDJ} has nonnegative coefficients.

Proof. The numerator of F_5^{GDJ} factors into thirteen simple factors. One of them, $1 - 2x_2 x_3$, cancels a denominator factor. The rest group as displayed. Both identities are checked exactly as polynomial identities after clearing denominators (`chernpp.ballot.d5_positive_form_checks`). Each V_i has the form $v/(1 - u - v)$ or $u/(1 - 2u)$ with u, v monomials, and every term of $\mathcal{C}$ is a monomial over products of factors $1 - 2m$ and $1 - m - m'$. $\square$

The eleven-term decomposition of $\mathcal{C}$ is [3, (16)].

Remark 13 (a second block, and why the $d = 4$ argument does not transfer). The factors can also be paired the other way: $(1 - p)/(1 - 2p)$ among the ratios, and

$$\frac{(1 - x_1)(1 - x_1 x_2)(1 - r)(1 - q)(1 - r - 2p)(1 - q - 2p)}{(1 - 2x_1)(1 - 2x_1 x_2)(1 - q - p)(1 - r - p)(1 - 2q)(1 - r - q)}$$

as the block. Its numerator is the same as that of $\mathcal{C}$: the product of the numerators of two copies of the $d = 4$ block of Lemma 8. It has an exact certificate $\sum_S P_S \prod_{r \in S} (1 - f_r) + P_0$ with 17 nonnegative rational parts over subsets of at most five factors (`results/d5_block_certificate.json`, re-verified in exact rationals by `chernpp.ballot.verify_d5_block`), a second proof that $F_5^{GDJ} \geq 0$. The degree-3 projection of that linear programme is infeasible at every order ≤ 4 . This rules out certificates of lower order at every degree, modulo the LP solver's infeasibility verdict (`chernpp.ballot.d5_alternative_block`).

The certificate's weights are fractional, and no integer certificate of the same shape exists with parts of degree ≤ 10 : the integer programme is infeasible while its relaxation is not. So Step 1 of the $d = 4$ argument, which counts supports and needs every term to contribute whole units, does not transfer. The endgame below avoids it. It needs only $F_5^{GDJ} \geq 0$ and lower bounds at single cells, and those come from the integer decomposition of $\mathcal{C}$.

5.4 The theorem

Theorem 14. *For every zero-sum multiset M of five integers, $C(M) \geq b(M)$, with equality if and only if $\max M \leq 1$. Moreover $C(M) \geq b(M) + 1$ whenever $\max M \geq 2$.*

Proof. By Proposition 11 and Lemma 12, $C(M)$ is a sum of nonnegative coefficients of F_5^{GDJ} , one per ballot ordering. The decreasing ordering is ballot, so $C(M) \geq F_5^{GDJ}(\beta_{\max}(M))$.

Small charge. There are finitely many M with $\max M \leq 9$, namely 2611. All of them are complete in the charge box $p = 9$. The exact table (`chernpp.bboxes`, reconstructed by CRT and checked against a held-back prime) gives $C = b$ on the 7 plane-sector packets and $C \geq b + 1$ on the other 2604.

Large charge. Every factor in Lemma 12 is a sum over explicit parameters of explicit coefficients. The coefficient of $1/(1 - 2m)$ at m^k is 2^k , and the coefficient of $1/(1 - m - m')$ at $m^j m'^n$ is $\binom{j+n}{n} \geq 2^{\min(j,n)}$. Expanding $C \cdot \prod_i (1 + V_i)$ gives $12 \cdot 64$ product terms with nonnegative coefficients. The coefficient of a product term at β is at least the product of the factors' coefficients along any one way of writing β as a sum of their exponents. So the set of β at which some product term is $\geq 2^7$ is a Presburger set.

isl decides that it contains every cell of the decreasing cone $\{\beta : \alpha_1 \geq \dots \geq \alpha_5, \beta_1 \geq 10\}$ (`chernpp.ballot.verify_d5`). Twelve product terms suffice. Eleven of them are products of the V_i alone; the twelfth is $\frac{x_1}{A} V_1 V_5 V_6$. Hence for $\max M \geq 10$, $C(M) \geq F_5^{GDJ}(\beta_{\max}(M)) \geq 128 > 5! \geq b(M)$. $\square$

Remark 15. The same argument re-proves Theorem 9 independently of its Steps 2 and 3. By Lemma 8, $F_4 \geq 0$, and isl shows $F_4(\beta_{\max}(M)) \geq 2^5 > 4!$ for $\max M \geq 9$. The 351 packets with $\max M \leq 8$ are checked from the exact table (`chernpp.ballot.verify_d4_dominant`). The atomic coefficient claims behind both isl computations are checked against the true series in tier 15. Growth on the decreasing cone comes almost entirely from the Lemma-1 ratios, not from the block.

6 Zero insertion

Appending a 0 to M adds a singleton block and multiplies the Chern monomial by $c_{\ell+1}$. For a zero-free base M_0 write $C_z = C(M_0 \sqcup 0^z)$ and $b_z = b(M_0 \sqcup 0^z)$. On the plane sector b_z is a Kreweras number, a polynomial in z . For example $b(1, -1, 0^z) = \binom{z+2}{2} = 1 + 2z + \binom{z}{2}$.

Conjecture 16 (zero insertion). *For every zero-free M_0 , the sequences C_z and $C_z - b_z$ have nonnegative forward differences of every order at $z = 0$:*

$$C_z - b_z = \sum_{k \geq 0} N_k(M_0) \binom{z}{k}, \quad N_k(M_0) \geq 0.$$

This refines Conjecture 5, since every M is some $M_0 \sqcup 0^z$. It holds on all 10,620 zero-free bases available (`results/zero_insertion.json`), for C and for $C - b$, at every order that fits in $d \leq 7$. Where the differences terminate within range, C_z is a polynomial:

$$C(2, -2, 0^z) = 2 + 5z + 4 \binom{z}{2} + \binom{z}{3}, \quad C(3, -3, 0^z) = 4 + 13z + 16 \binom{z}{2} + 9 \binom{z}{3} + 2 \binom{z}{4}.$$

The degree equals $\sum_{a>0} a + \#\{a < 0\}$ in all four certified cases. That is too few to conjecture.

7 Where a proof has to go

In the geometric model of §2.1, a packet of charge p reduces by (I3) only to the curvilinear locus $C_{n,p+1}$ in $\mathbb{C}^{p+1}$. The plane case $p = 1$ is Theorem 2. Conjecture 5 therefore says:

the integral of $m_\nu(E)$ over the curvilinear locus in $\mathbb{C}^{p+1}$ is at least the count of ballot words, which is what the same integral gives in the plane, and exceeds it as soon as $p \geq 2$.

Two routes suggest themselves.

1. *A degeneration.* Degenerate $C_{n,p+1}$ to a cycle containing a planar component with multiplicity one, plus effective components on which $m_\nu(\epsilon)$ integrates nonnegatively. Positivity does fail on some individual degeneration components [3], so the decomposition would have to be chosen with care.
2. *A charge-2 Jucys–Murphy calculus* (the naive form is ruled out). Since symmetric functions of the J_k are central, (I4) says that on the plane

$$\int_{C_{n,2}} f(\epsilon) = \sum_{\lambda \vdash n} \frac{\dim \lambda \chi^\lambda(\sigma)}{n!} f(\text{contents}(\lambda) \setminus \{0\}),$$

a sum over hooks. The exact tables determine the whole functional $f \mapsto \int_{C_{n,3}} f(\epsilon)$ on symmetric functions of degree $2d$: every packet with $\max M \leq 2$ reduces to $r = 3$ by (I3). No weights w_λ , $\lambda \vdash n$, make it a content evaluation of the same shape: the linear system (7, 15, 30 equations against 5, 7, 11 unknowns at $d = 3, 4, 5$) is inconsistent (`tools/content_functional.py`). So the curvilinear locus in $\mathbb{C}^3$ is not governed by any central functional of the Jucys–Murphy elements of S_n . A charge-2 calculus needs a larger algebra.

3. *A charge-2 Jucys–Murphy calculus (what remains).* Find the analogue, for the curvilinear component of the punctual Hilbert scheme of $\mathbb{C}^3$, of the Lehn–Sorger description. In such a calculus the excess $C(M) - b(M)$ should count a positive family of *non-monotone* or *branched* factorisations.

The exact tables (`results/morin_a{6,7}.json`, 20,725 coefficients) are available to test either route.

A reduction at fixed d . The proof of Theorem 14 isolates what a proof at a fixed $d \geq 6$ needs. Suppose a residue-null kernel K is found, with nullity certified as in Proposition 11, and the chamber series of $\mathcal{Q}_d - K$ has a manifestly nonnegative product form. Then $C(M)$ is at least one coefficient of that series, at the decreasing cell of M . Wherever the product form grows exponentially along the decreasing cone, that coefficient exceeds $d! \geq b(M)$, and what remains is a finite table. At $d = 5$ the growth comes almost entirely from the Lemma-1 ratios. The only missing ingredient at $d = 6, 7$ is therefore a nonnegative gauge. None is known: at $d = 6$ the structural kernel families of the project’s gauge search suppress every negative coefficient to a finite depth but not beyond [8].

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